# Supplementary Tables

*Affective Information Transmission in Cross-Species Acoustic Communication: Evidence for Multiscale Acoustic–Affective Organization*

## Supplementary Table S1

**Pairwise Euclidean distances between acoustic-category markers in the Figure 5A LNU1–LNU2 representation.**

| Category A | Category B | Euclidean distance (LNU) |
|---|---|---|
| High-amplitude range shouts | Pink noise | 0.055 |
| Grazing calls (bovine) | Kulning herding call | 0.124 |
| Baseline (normalized noise floor) | High-amplitude range shouts | 0.221 |
| Cattle-herding calls (vaquería, voz) | Pre-slaughter holding calls (bovine) | 0.264 |
| Baseline (normalized noise floor) | Pink noise | 0.268 |
| Kulning herding call | Milking songs | 0.422 |
| Herding whistles | Pink noise | 0.485 |
| Herding whistles | High-amplitude range shouts | 0.507 |
| Grazing calls (bovine) | Milking songs | 0.545 |
| Baseline (normalized noise floor) | Pre-slaughter holding calls (bovine) | 0.652 |
| Baseline (normalized noise floor) | Herding whistles | 0.708 |
| High-amplitude range shouts | Pre-slaughter holding calls (bovine) | 0.845 |
| Pink noise | Pre-slaughter holding calls (bovine) | 0.875 |
| Baseline (normalized noise floor) | Cattle-herding calls (vaquería, voz) | 0.914 |
| Cattle-herding calls (vaquería, voz) | High-amplitude range shouts | 1.109 |
| Cattle-herding calls (vaquería, voz) | Pink noise | 1.138 |
| Herding whistles | Pre-slaughter holding calls (bovine) | 1.351 |
| Cattle-herding calls (vaquería, voz) | Herding whistles | 1.616 |

## Supplementary Table S1 (continued)

| Category A | Category B | Euclidean distance (LNU) |
|---|---|---|
| Herding whistles | Milking songs | 1.960 |
| Herding whistles | Kulning herding call | 2.380 |
| Milking songs | Pink noise | 2.445 |
| High-amplitude range shouts | Milking songs | 2.464 |
| Grazing calls (bovine) | Herding whistles | 2.503 |
| Baseline (normalized noise floor) | Milking songs | 2.648 |
| Kulning herding call | Pink noise | 2.864 |
| High-amplitude range shouts | Kulning herding call | 2.882 |
| Grazing calls (bovine) | Pink noise | 2.988 |
| Grazing calls (bovine) | High-amplitude range shouts | 3.007 |
| Baseline (normalized noise floor) | Kulning herding call | 3.064 |
| Baseline (normalized noise floor) | Grazing calls (bovine) | 3.188 |
| Milking songs | Pre-slaughter holding calls (bovine) | 3.300 |
| Cattle-herding calls (vaquería, voz) | Milking songs | 3.562 |
| Kulning herding call | Pre-slaughter holding calls (bovine) | 3.715 |
| Grazing calls (bovine) | Pre-slaughter holding calls (bovine) | 3.839 |
| Cattle-herding calls (vaquería, voz) | Kulning herding call | 3.977 |
| Cattle-herding calls (vaquería, voz) | Grazing calls (bovine) | 4.101 |

**Note.** Distances are Euclidean distances between category-wise median centroid coordinates in the LNU1–LNU2 plane underlying Figure 5A. Small display offsets applied to numbered symbols for legibility are not included in these calculations. All 36 unordered category pairs are listed and rounded to three decimals. Smaller values indicate closer positions in this displayed two-dimensional representation; they do not establish acoustic, affective, or semantic equivalence.

## Supplementary Table S2

**Descriptive source-directed orientation across acoustic categories during pre-compositional playback.**

| Acoustic category | N (one-minute analytical intervals) | Mean source-directed orientation (s/min-equivalent) | SD |
|---|---|---|---|
| Baseline (normalized noise floor) | 44 | 14.0 | 4.3 |
| Pink noise | 14 | 12.6 | 4.5 |
| Herding whistles | 4 | 11.4 | 3.9 |
| Grazing calls (bovine) | 8 | 17.6 | 1.4 |
| Milking songs | 4 | 16.5 | 2.2 |
| Kulning herding call | 4 | 17.1 | 2.2 |
| High-amplitude range shouts | 4 | 11.5 | 4.1 |
| Cattle-herding calls (vaquería, voz) | 2 | 9.0 | 0.6 |
| Pre-slaughter holding calls (bovine) | 6 | 10.8 | 3.1 |

**Note.** Baseline denotes rest/silence intervals in the playback masters, labelled ‘Baseline (rest, normalized noise floor)’ in the source data; pink noise is a separate stimulus condition. Values summarize the proportion of cattle displaying source-oriented ethogram configurations in each one-minute group observation. For presentation, the proportion was multiplied by 60 s/min and is reported on an s/min-equivalent scale; it is not the duration of an individual animal’s gaze. The dataset contains 90 intervals: 30 observations from each of three assembled playback masters (T1–T3). N is the number of repeated analytical intervals, not the number of animals or independent biological replicates. The source-oriented proportion used a scan-specific denominator derived as the sum of four distance-zone counts; these denominators ranged from 68 to 72. Values are the mean and standard deviation across intervals in each category. These summaries are descriptive and do not imply preference, attraction, positive affect, or welfare benefit.